

DSC 140B

Representation Learning

Lecture 04 | Part 1

News

News

- ▶ Quiz 02 tonight.
- ▶ See Campuswire post re: Peter Chi's study on data science education.
 - ▶ **You get a \$10 gift card to the bookstore.**

▷ Waitlist

Live QA

Exercise

Are you planning on taking the quiz tonight?

True = Yes

False = No

DSC 140B

Representation Learning

Lecture 04 | Part 2

The Spectral Theorem

Eigenvectors

- Let A be an $n \times n$ matrix. An **eigenvector** of A with **eigenvalue** λ is a nonzero vector $\vec{v}$ such that $A\vec{v} = \lambda\vec{v}$.

Eigenvectors (of Linear Transformations)

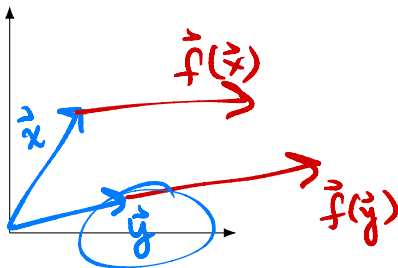
- Let $\vec{f}$ be a linear transformation. An **eigenvector** of $\vec{f}$ with **eigenvalue** λ is a nonzero vector $\vec{v}$ such that $\vec{f}(\vec{v}) = \lambda\vec{v}$.

Importance

- ▶ We will see why eigenvectors are important in the next part.
- ▶ For now: what are they?

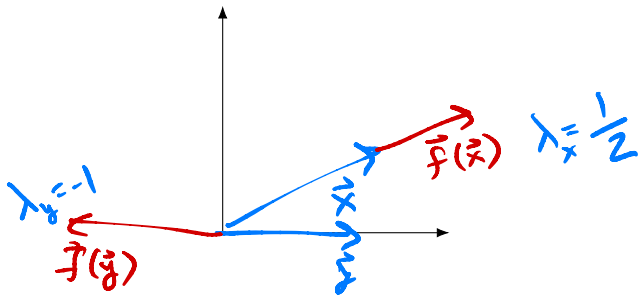
Geometric Interpretation

- Recall: $\vec{v}$ is an **eigenvector** if $\vec{f}(\vec{v}) = \lambda\vec{v}$.
- Meaning: when $\vec{f}$ is applied to one of its eigenvectors, $\vec{f}$ simply scales it.



Geometric Interpretation

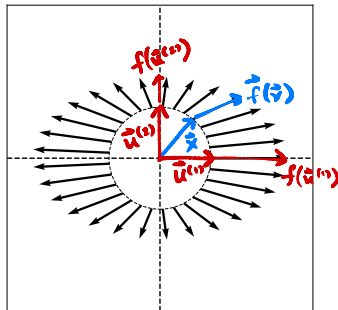
- ▶ The **eigenvalue**, λ , tells us how much the eigenvector is scaled.
 - ▶ If $\lambda > 1$, the eigenvector is stretched.
 - ▶ If $0 < \lambda < 1$, the eigenvector is shrunk.
 - ▶ If $\lambda < 0$, the eigenvector is flipped and scaled.



Exercise

Draw as many (linearly independent) eigenvectors as you can.

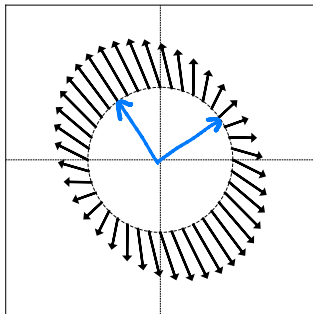
$$A = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}$$



Exercise

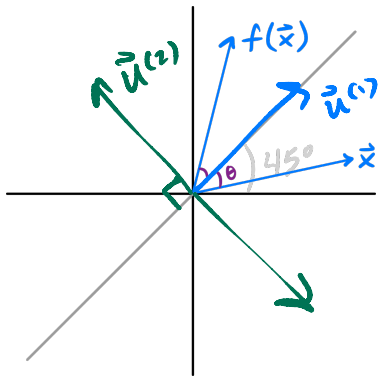
Draw as many (linearly independent) eigenvectors as you can.

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$$



Exercise

Consider the linear transformation which mirrors its input over the line of 45° . Give two orthogonal eigenvectors of the transformation.



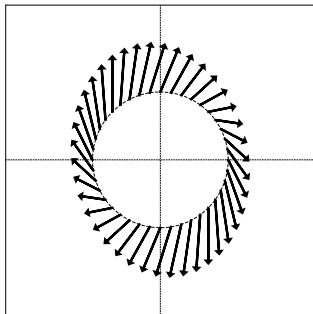
$$\begin{aligned} \vec{f}(\vec{u}^{(1)}) &= \vec{u}^{(1)} & \lambda_1 &= 1 \\ \vec{f}(\vec{u}^{(2)}) &= -\vec{u}^{(2)} & \lambda_2 &= -1 \end{aligned}$$

Exercise

Draw as many (linearly independent) eigenvectors as you can.

None

$$A = \begin{pmatrix} 5 & 5 \\ -10 & 12 \end{pmatrix}$$

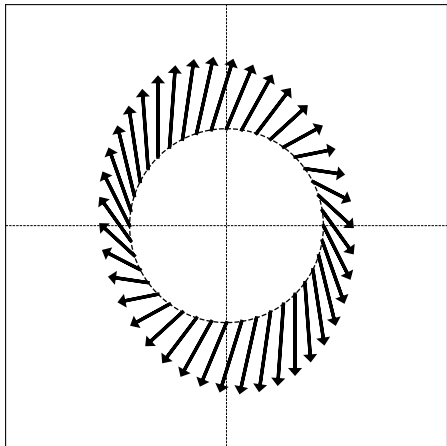


Caution!

- ▶ Not all matrices have even one eigenvector!¹
- ▶ When does a matrix have multiple (linearly independent) eigenvectors?

¹That is, with a *real-valued* eigenvalue.

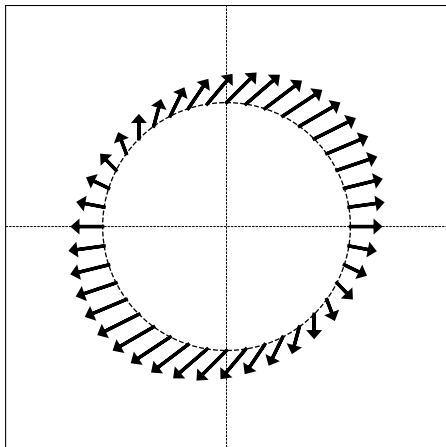
Example Linear Transformation



Name

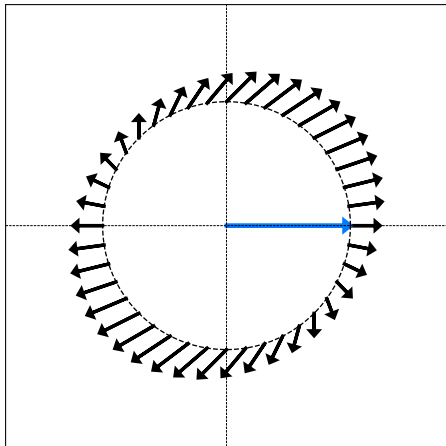
$$A = \begin{pmatrix} 5 & 5 \\ -10 & 12 \end{pmatrix}$$

Example Linear Transformation



$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

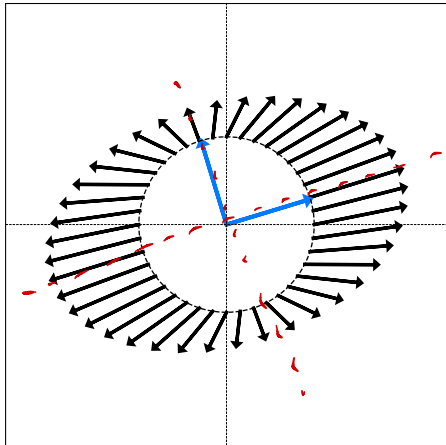
Example Linear Transformation



$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

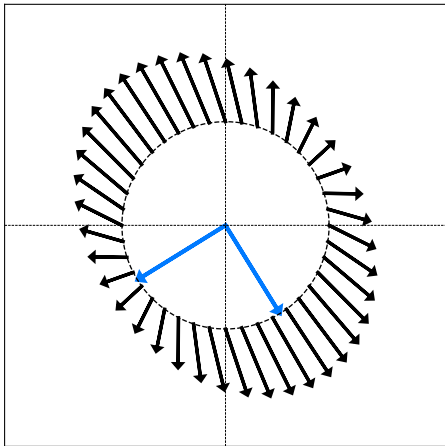
$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

Example **Symmetric** Linear Transformation



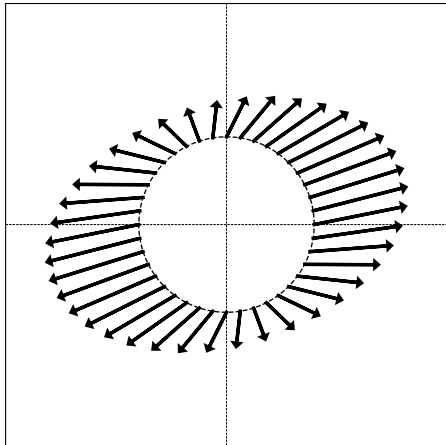
$$A = \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$$

Example **Symmetric** Linear Transformation



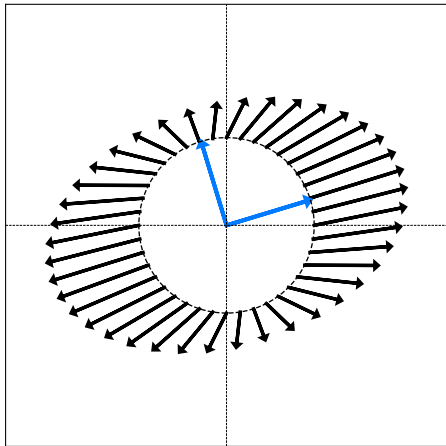
$$A = \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$$

Example **Symmetric** Linear Transformation



$$A = \begin{pmatrix} 5 & 1 \\ 1 & 2 \end{pmatrix}$$

Example **Symmetric** Linear Transformation



$$A = \begin{pmatrix} 5 & 1 \\ 1 & 2 \end{pmatrix}$$

Observation

- ▶ It seems that there is something special about **symmetric** matrices...

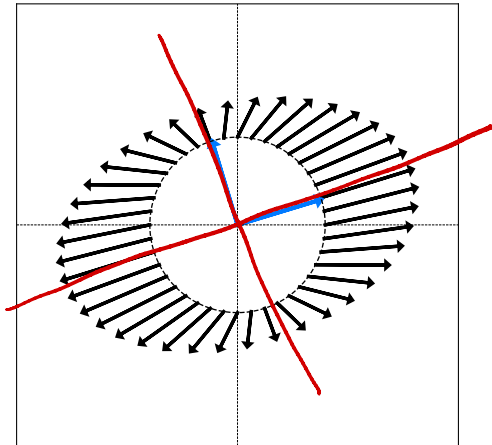
Symmetric Matrices

- ▶ Recall: a matrix A is **symmetric** if $A^T = A$.

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}$$

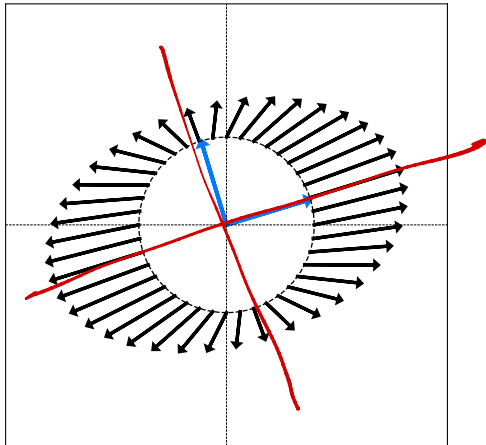
- ▶ A linear transformation $\vec{f}$ is **symmetric** if its matrix representation is symmetric.

Observation #1



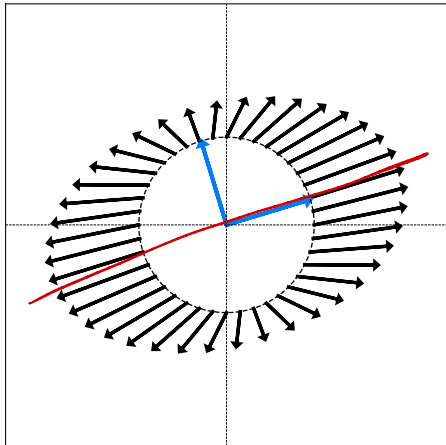
- Symmetric linear transformations have **axes of symmetry**.
 - One for each dimension.

Observation #2



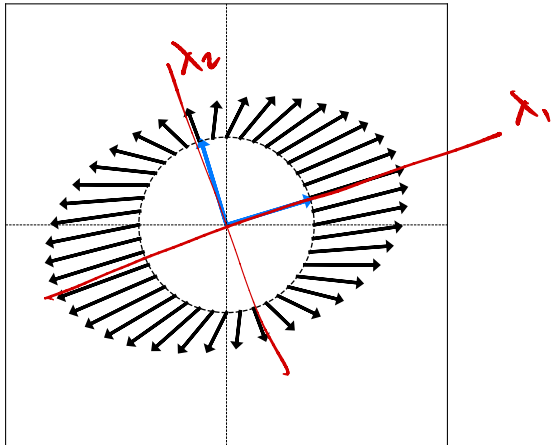
- The axes of symmetry are **orthogonal** to one another.

Observation #3



- ▶ The action of $\vec{f}$ along an axis of symmetry is simply to **scale** its input.
- ▶ That is, the **eigenvectors** point along the axes of symmetry.

Observation #4



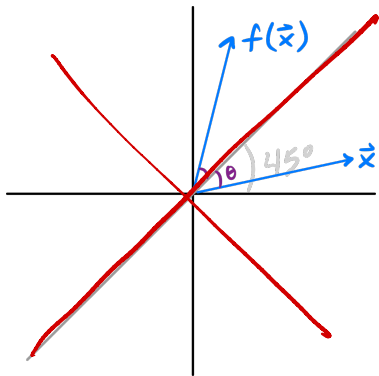
- The size of this scaling can be different for each axis.

Main Idea

The **eigenvectors** of a symmetric linear transformation (matrix) are its axes of symmetry. The **eigenvalues** describe how much each axis of symmetry is scaled.

Exercise

Consider the linear transformation which mirrors its input over the line of 45° . Give two orthogonal eigenvectors of the transformation.



How many?

- ▶ The symmetric 2×2 matrices we saw all had 2 orthogonal eigenvectors.
- ▶ Does a 3×3 symmetric matrix have 3 orthogonal eigenvectors?
- ▶ What about $n \times n$ symmetric matrices?

The Spectral Theorem²

Theorem

Let A be an $n \times n$ **symmetric** matrix. Then you can always find n eigenvectors of A which are all mutually orthogonal.

²for symmetric matrices

Careful!

- ▶ The spectral theorem *does not* say that an $n \times n$ matrix has n eigenvectors!

Live Q&A

Exercise

Consider the 2×2 identity matrix. How many (unit) eigenvectors does it have?

A. 0

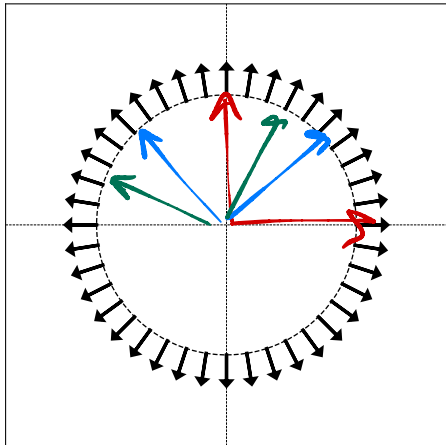
B. 1

C. 2

D. ∞

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Solution



- ▶ Infinitely many!
- ▶ *Every* (nonzero) vector is an eigenvector with eigenvalue 1.

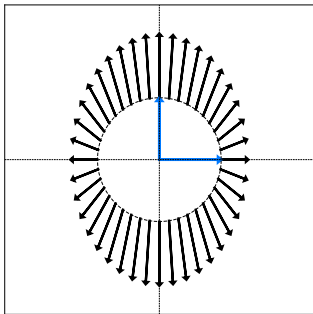
Solution

- ▶ It would be incorrect to say that the identity matrix has just 2 orthogonal eigenvectors.
- ▶ Instead, the spectral theorem says: “You can find 2 different orthogonal eigenvectors of I .”
- ▶ There are infinitely-many ways to do this!
 - ▶ $(1, 0)^T$ and $(0, 1)^T$
 - ▶ $(1/\sqrt{2}, 1/\sqrt{2})^T$ and $(-1/\sqrt{2}, 1/\sqrt{2})^T$
 - ▶ etc.

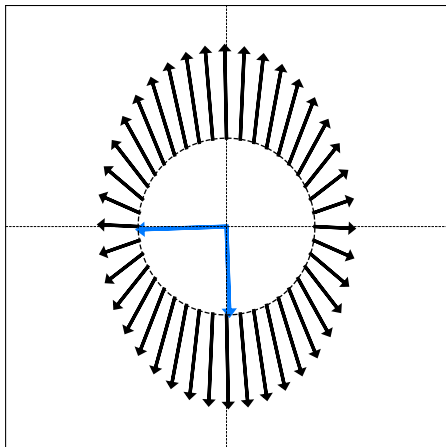
Diagonal Matrices

- If A is diagonal, its eigenvectors are simply the standard basis vectors.

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}$$

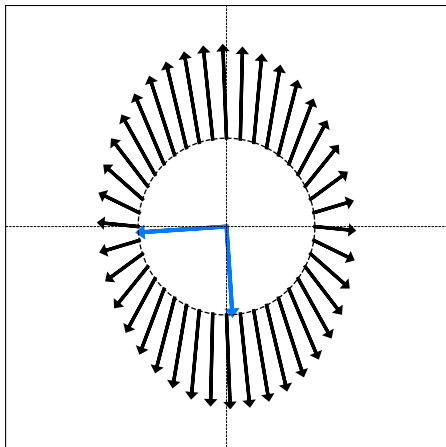


Off-diagonal elements



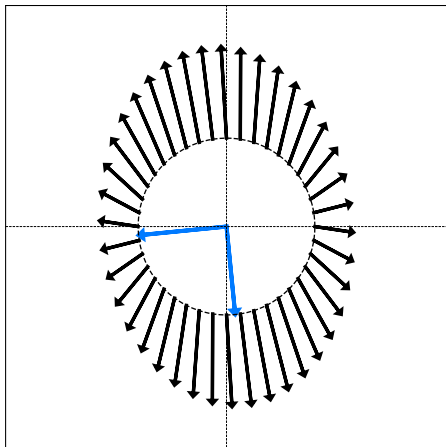
$$A = \begin{pmatrix} 2 & -0.1 \\ -0.1 & 5 \end{pmatrix}$$

Off-diagonal elements



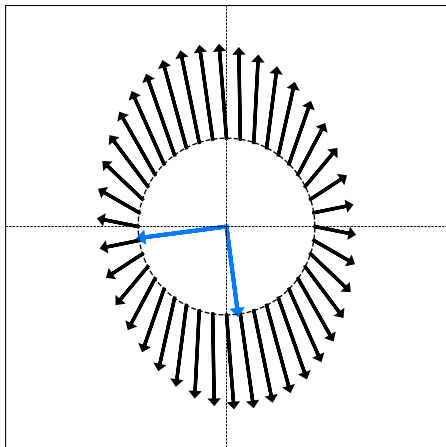
$$A = \begin{pmatrix} 2 & -0.2 \\ -0.2 & 5 \end{pmatrix}$$

Off-diagonal elements



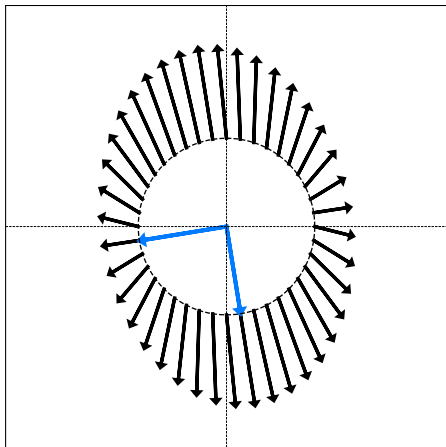
$$A = \begin{pmatrix} 2 & -0.3 \\ -0.3 & 5 \end{pmatrix}$$

Off-diagonal elements



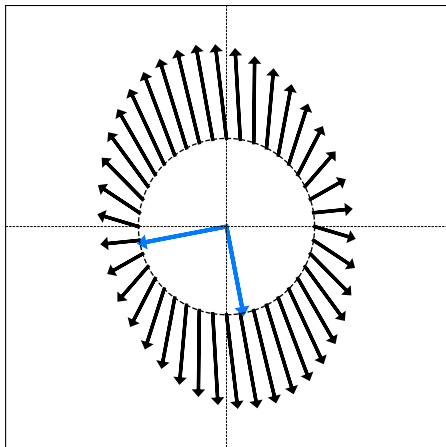
$$A = \begin{pmatrix} 2 & -0.4 \\ -0.4 & 5 \end{pmatrix}$$

Off-diagonal elements



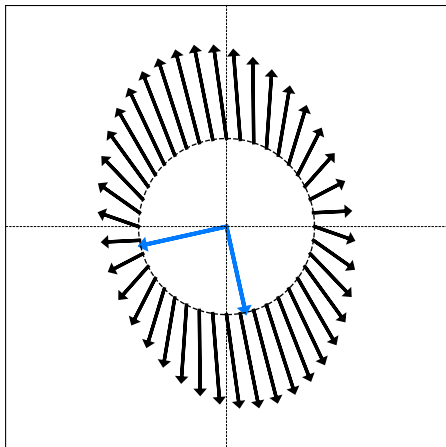
$$A = \begin{pmatrix} 2 & -0.5 \\ -0.5 & 5 \end{pmatrix}$$

Off-diagonal elements



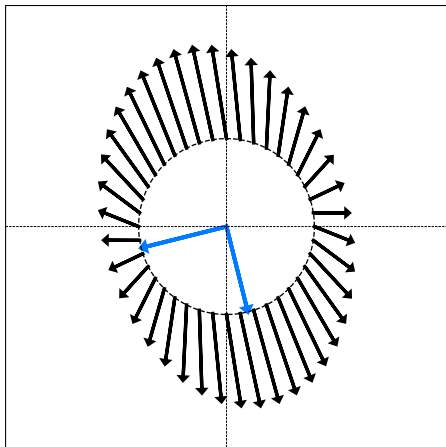
$$A = \begin{pmatrix} 2 & -0.6 \\ -0.6 & 5 \end{pmatrix}$$

Off-diagonal elements



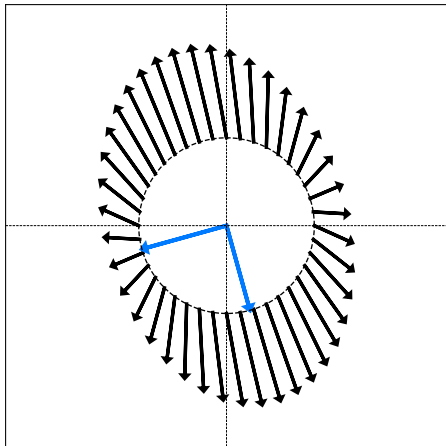
$$A = \begin{pmatrix} 2 & -0.7 \\ -0.7 & 5 \end{pmatrix}$$

Off-diagonal elements



$$A = \begin{pmatrix} 2 & -0.8 \\ -0.8 & 5 \end{pmatrix}$$

Off-diagonal elements



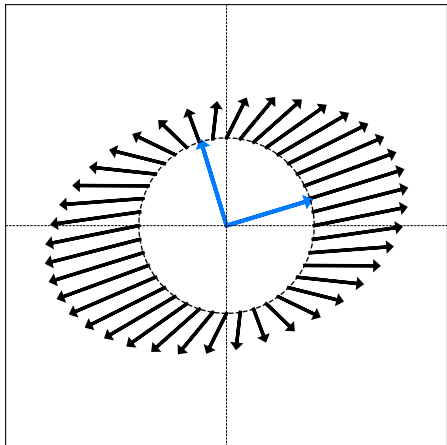
$$A = \begin{pmatrix} 2 & -0.9 \\ -0.9 & 5 \end{pmatrix}$$

Non-Diagonal Symmetric Matrices

- ▶ When a symmetric matrix is not diagonal, its eigenvectors are not the standard basis vectors.
- ▶ But they are still orthogonal!

Computing Eigenvectors

$$\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$$



Use `np.linalg.eigh`^a:

```
>> A = np.array([[2, -1], [-1, 3]])  
>> np.linalg.eigh(A)  
(array([1.38196601, 3.61803399]),  
 array([[ -0.85065081, -0.52573111],  
        [ -0.52573111,  0.85065081]]))
```

^aif the input is *symmetric*

DSC 140B

Representation Learning

Lecture 04 | Part 3

Why are eigenvectors useful?

OK, but why are eigenvectors³ useful?

1. Eigenvectors are natural **basis vectors**.
2. Eigenvectors are **equilibria**.
3. Eigenvectors are **maximizers** (or minimizers).

³of symmetric matrices

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Recall: Spectral Theorem

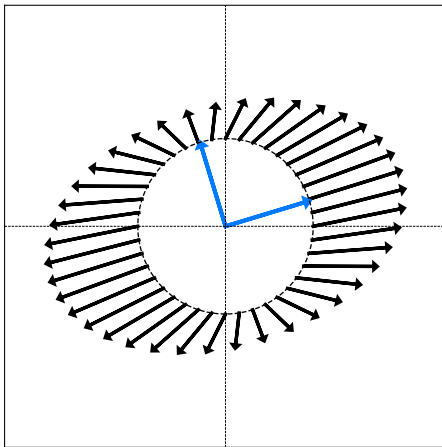
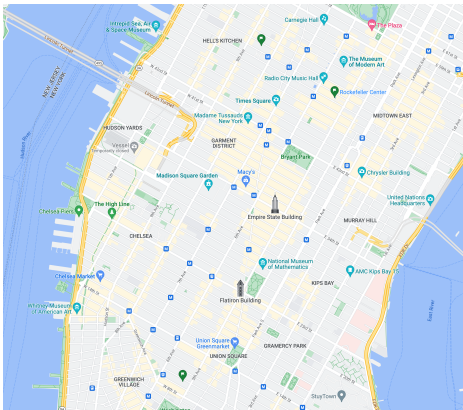
Theorem

Let A be a symmetric $d \times d$ matrix. Then you can find d orthonormal eigenvectors $\hat{u}^{(1)}, \dots, \hat{u}^{(d)}$ of A .

- In other words, you can make an orthonormal basis out of eigenvectors of A .



“Nice” Bases



Using the Eigenbasis

- ▶ When we work in the **eigenbasis** of A , many things become simpler.

Example

- ▶ Consider the symmetric matrix A .
- ▶ If we change basis, A changes.
- ▶ What does it look like if we change to the eigenbasis of A ?

$$A = \begin{pmatrix} 4 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 4 \end{pmatrix}$$


Hand-drawn blue arrows indicate the mapping of the columns of matrix A to eigenvectors:

- A vertical arrow points from the first column (4, 2, 1) to the label $f(\hat{e}^{(1)})$.
- A curved arrow points from the second column (2, 3, 2) to the label $f(\hat{e}^{(2)})$.
- A curved arrow points from the third column (1, 2, 4) to the label $f(\hat{e}^{(3)})$.

Example

- ▶ Consider the symmetric matrix A .
- ▶ If we change basis, A changes.
- ▶ What does it look like if we change to the eigenbasis of A ?

$$A = \begin{pmatrix} 4 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 4 \end{pmatrix} \xrightarrow{\text{eigenbasis}} [A]_{\mathcal{U}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 7 \end{pmatrix}$$

- ▶ It becomes diagonal!

Example

- ▶ Evaluating the linear transformation becomes easier, too.
- ▶ Suppose $\vec{x} = (3, 2, 1)^T$. Before:

$$\vec{f}(\vec{x}) = A\vec{x}$$

$$= \begin{pmatrix} 4 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 4 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \times 3 + 2 \times 2 + 1 \times 1 \\ 2 \times 3 + 3 \times 2 + 2 \times 1 \\ 1 \times 3 + 2 \times 2 + 4 \times 1 \end{pmatrix} = \begin{pmatrix} 17 \\ 14 \\ 11 \end{pmatrix}$$

Example

- In the eigenbasis, $\vec{x}$'s coordinates are:

$$[\vec{x}]_{\mathcal{U}} = (0, \sqrt{2}, 2\sqrt{3})^T$$

- So:

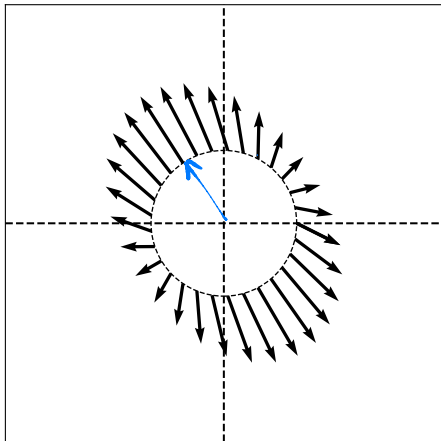
$$[A]_{\mathcal{U}}[\vec{x}]_{\mathcal{U}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 7 \end{pmatrix} \begin{pmatrix} 0 \\ \sqrt{2} \\ 2\sqrt{3} \end{pmatrix} = \begin{pmatrix} 0 \\ 3\sqrt{2} \\ 14\sqrt{3} \end{pmatrix}$$

OK, but why are eigenvectors⁴ useful?

1. Eigenvectors are natural **basis vectors**.
2. Eigenvectors are **equilibria**.
3. Eigenvectors are **maximizers** (or minimizers).

⁴of symmetric matrices

Eigenvectors are Equilibria



- ▶ $\vec{f}(\vec{x})$ rotates $\vec{x}$ towards the “top” eigenvector $\vec{v}$.
- ▶ $\vec{v}$ is an equilibrium.

Use Case: The Power Method

- ▶ Method for computing the top eigenvector/value of A .
- ▶ Initialize $\vec{x}^{(0)}$ randomly
- ▶ Repeat until convergence:
 - ▶ Set $\vec{x}^{(i+1)} = A\vec{x}^{(i)} / \|A\vec{x}^{(i)}\|$

OK, but why are eigenvectors⁵ useful?

1. Eigenvectors are natural **basis vectors**.
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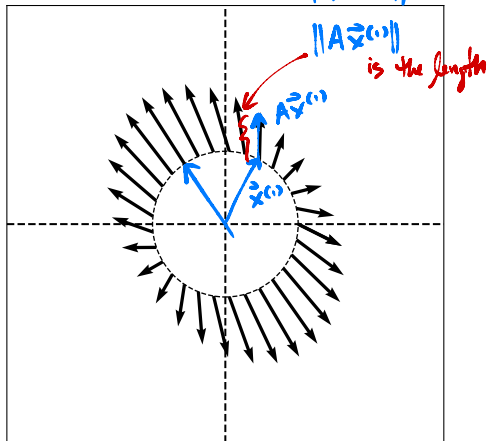
Eigenvectors as Optimizers

- ▶ Eigenvectors are the solutions to certain common optimization problems involving matrices / linear transformations.
- ▶ This might be **the** most important reason why eigenvectors are useful in **data science**.

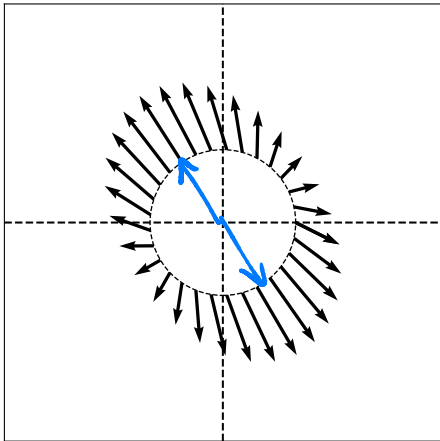
Exercise

Draw a unit vector $\vec{x}$ such that $\|A\vec{x}\|$ is largest.

$\|\vec{f}(\vec{x})\|$



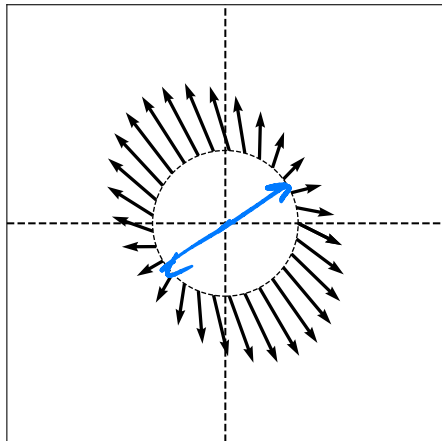
Observation #1



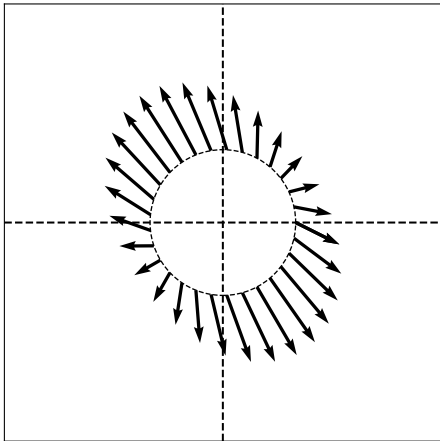
- ▶ $\vec{f}(\vec{x})$ is longest along the “main” axis of symmetry.
 - ▶ In the direction of the eigenvector with largest eigenvalue.

Exercise

Draw a unit vector $\vec{x}$ such that $\|A\vec{x}\|$ is smallest.



Observation #2



- ▶ $\vec{f}(\vec{x})$ is smallest along the “minor” axis of symmetry.
 - ▶ In the direction of the eigenvector with smallest eigenvalue.

Main Idea

Suppose A is a symmetric matrix.

To maximize $\|A\vec{x}\|$ over unit vectors, pick $\vec{x}$ to be a top eigenvector of A . That is, an eigenvector with the largest eigenvalue (in abs. value).

To minimize $\|A\vec{x}\|$, pick $\vec{x}$ to be a bottom eigenvector. That is, an eigenvector with the smallest eigenvalue (in abs. value).

Main Idea

Suppose $\vec{f}$ is a symmetric linear transformation.

To maximize $\|\vec{f}(\vec{x})\|$ over unit vectors, pick $\vec{x}$ to be a top eigenvector of $\vec{f}$.

To minimize $\|\vec{f}(\vec{x})\|$ over unit vectors, pick $\vec{x}$ to be a bottom eigenvector.

Also true...

- $\vec{x} \cdot A\vec{x}$ is called a **quadratic form**.

Theorem

Let A be a symmetric matrix.

To maximize $\vec{x} \cdot A\vec{x}$ over unit vectors, pick $\vec{x}$ to be a top eigenvector of A .

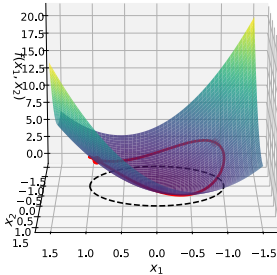
To minimize $\vec{x} \cdot A\vec{x}$ over unit vectors, pick $\vec{x}$ to be a bottom eigenvector of A .

By the way...

- ▶ We'll walk you through the proofs in the homework.

Example

- **Problem:** Maximize $f(x_1, x_2) = 4x_1^2 + 2x_2^2 + 3x_1x_2$
subject to $x_1^2 + x_2^2 = 1$



Solution

- ▶ **Problem:** Maximize $f(x_1, x_2) = 4x_1^2 + 2x_2^2 + 3x_1x_2$
subject to $x_1^2 + x_2^2 = 1$

- ▶ You can write $f(x_1, x_2)$ as $f(\vec{x}) = \vec{x} \cdot A\vec{x}$ where

$$A = \begin{pmatrix} 4 & 1.5 \\ 1.5 & 2 \end{pmatrix} \quad \vec{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

- ▶ Top eigenvector of A is approximately:
 $(0.88, 0.47)^T$
- ▶ Solution: maximized at $x_1 = 0.88, x_2 = 0.47$

Next time...

- ▶ Change of basis matrices, diagonalization.
- ▶ Dimensionality reduction (**actual ML!**)